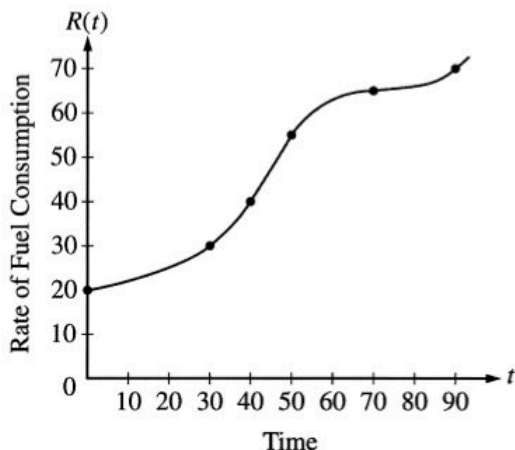


8-1b

1.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane.

Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part D

1 point is earned for the meaning of $\int_0^b R(t) dt$

$\int_0^b R(t) dt$ is the total amount of fuel in gallons consumed for the first b minutes.

< -1 > if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t) dt$

$\frac{1}{b} \int_0^b R(t) dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

< -1 > if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
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The student response earns three of the following points:

1 point is earned for the meaning of $\int_0^b R(t) dt$

$\int_0^b R(t) dt$ is the total amount of fuel in gallons consumed for the first b minutes.

8-1b

< -1 > if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t) dt$

$\frac{1}{b} \int_0^b R(t) dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

< -1 > if no reference to time b

1 point is earned for the correct units in both answers

8-1b

2. Let f be a function defined by $f(x) = \begin{cases} 1 - 2 \sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

(a) Show that f is continuous at $x = 0$.

(b) For $x \neq 0$, express $f'(x)$ as a piecewise-defined function. Find the value of x for which $f'(x) = -3$.

(c) Find the average value of f on the interval $[-1, 1]$.

Part A

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.

Part B

8-1b

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2 \cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2 \cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right) > 0.$$

$$\text{Therefore, } f'(x) = -3 \text{ for } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right).$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2 \cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2 \cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right) > 0.$$

$$\text{Therefore, } f'(x) = -3 \text{ for } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right).$$

8-1b

Part C

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

8-1b

$$\begin{aligned}\int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\&= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\&= [x + 2 \cos]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\&= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right)\end{aligned}$$

$$\begin{aligned}\text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\&= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4}\end{aligned}$$

8-1b

3. Let f be a function defined by $f(x) = \begin{cases} 1 - 2 \sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

Find the average value of f on the interval $[-1, 1]$.

Part C

The student response earns none of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4} e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4} e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8} e^{-4} \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

8-1b

$$\begin{aligned}\int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\&= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\&= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\&= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right)\end{aligned}$$

$$\begin{aligned}\text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\&= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4}\end{aligned}$$

8-1b

4. Let f be the function given by $f(x) = x^3 - 6x^2 + p$, where p is an arbitrary constant.
- (a) Write an expression for $f'(x)$ and use it to find the relative maximum and minimum values of f in terms of p . Show the analysis that leads to your conclusion.
- (b) For what values of the constant p does f have 3 distinct roots?
- (c) Find the value of p such that the average value of f over the closed interval $[-1, 2]$ is 1.

Part A

1 point is earned for correctly finding $f'(x)$

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving $f'(x) = 0$

$$3x(x - 4) = 0$$

$$x = 0, x = 4$$

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for p

$f'(x)$ changes sign from positive to negative at $x = 0$

$f'(x)$ changes sign from negative to positive at $x = 4$

or

$$f''(x) = 6x - 12, f''(0) = -12, f''(4) = 12$$

1 point is earned for correctly finding the maximum and minimum values in terms of p

relative maximum at $x = 0, f(0) = p$

relative minimum at $x = 4, f(4) = p - 32$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for correctly finding $f'(x)$

8-1b

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving $f'(x) = 0$

$$3x(x - 4) = 0$$

$$x = 0, x = 4$$

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for p

$f(x)$ changes sign from positive to negative at $x = 0$

$f(x)$ changes sign from negative to positive at $x = 4$

or

$$f''(x) = 6x - 12, f''(0) = -12, f''(4) = 12$$

1 point is earned for correctly finding the maximum and minimum values in terms of p

relative maximum at $x = 0, f(0) = p$

relative minimum at $x = 4, f(4) = p - 32$

Part B

1 point is earned for the correct upper bound

1 point is earned for the correct lower bound

$<-1>$ including 0 or 32, or both

$f(x)$ has three distinct real roots when $p > 0$ and $p - 32 < 0$,

so $0 < p < 32$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct upper bound

1 point is earned for the correct lower bound

8-1b

$<-1>$ including 0 or 32, or both

$f(x)$ has three distinct real roots when $p > 0$ and $p - 32 < 0$,

so $0 < p < 32$

Part C

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for p where the equation must be of the

form $k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$

$$\frac{1}{2 - (-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[\frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[\left(\frac{16}{4} - 16 + 2p \right) - \left(\frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for p where the equation must be of the

form $k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$

8-1b

$$\frac{1}{2-(-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[\frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[\left(\frac{16}{4} - 16 + 2p \right) - \left(\frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only

8-1b

5. Let f be the function given by $f(x) = x^3 - 6x^2 + p$, where p is an arbitrary constant.
Find the value of p such that the average value of f over the closed interval $[-1, 2]$ is 1.

Part C

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for p where the equation must be of the

$$\text{form } k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{2 - (-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[\frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[\left(\frac{16}{4} - 16 + 2p \right) - \left(\frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for p where the equation must be of the

$$\text{form } k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{2 - (-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[\frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[\left(\frac{16}{4} - 16 + 2p \right) - \left(\frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only

8-1b

8-1b

6. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t - 1)(t - 3)$. At time $t = 2$, the position of the particle is $x(2) = 0$.
- Find the minimum acceleration of the particle.
 - Find the total distance traveled by the particle.
 - Find the average velocity of the particle over the interval $0 \leq t \leq 5$.

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .

Part B

2 points are earned for integral expression

8-1b

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

8-1b

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$

Part C

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\begin{aligned}
 &\frac{\int_0^5 (3t^2 - 12t + 9)dt}{5 - 0} \\
 &= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5}(20) = 4
 \end{aligned}$$

or

8-1b

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\frac{\int_0^5 (3t^2 - 12t + 9) dt}{5-0}$$

$$= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5} (20) = 4$$

or

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$

8-1b

7. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t-1)(t-3)$. At time $t=2$, the position of the particle is $x(2)=0$.

Find the average velocity of the particle over the interval $0 \leq t \leq 5$

Part C

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1:

$$\frac{\int_0^5 (3t^2 - 12t + 9) dt}{5-0}$$

$$= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5} (20) = 4$$

Method 2:

$$\frac{x(5) - x(0)}{5-0} = \frac{18 - (-2)}{5} = 4$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1:

$$\frac{\int_0^5 (3t^2 - 12t + 9) dt}{5-0}$$

$$= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5} (20) = 4$$

Method 2:

$$\frac{x(5) - x(0)}{5-0} = \frac{18 - (-2)}{5} = 4$$

8-1b

8. Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.

(a) Find the area of R .

(b) If the line $x = k$ divides R into two regions of equal area, what is the value of k ?

(c) What is the average value of $y = \frac{x}{x^2+2}$ on the interval $0 \leq x \leq \sqrt{6}$?

Part A

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$

Part B

1 point is earned for a correct equation involving integral (s)

1 point is earned for antiderivative

8-1b

1 point is earned for evaluation of integral (s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \ln 2 &= \int_0^k \frac{x}{x^2 + 2} dx \\ &= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\ &= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2\end{aligned}$$

$$\therefore \frac{1}{2} \ln(k^2 + 2) = \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2$$

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for a correct equation involving integral (s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral (s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \ln 2 &= \int_0^k \frac{x}{x^2 + 2} dx \\ &= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\ &= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2\end{aligned}$$

$$\therefore \frac{1}{2} \ln(k^2 + 2) = \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2$$

8-1b

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$

Part C

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{x}{x^2 + 2} dx \\ &= \frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{x}{x^2 + 2} dx \\ &= \frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$

8-1b

9. Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.

What is the average value of $y = x / x^2 + 2$ on the interval $0 \leq x \leq \sqrt{6}$?

Part C

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{2}{x^2+2} dx \\ &= -\frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{2}{x^2+2} dx \\ &= -\frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$

8-1b

10. Two particles move along the x -axis. For $0 \leq t \leq 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.

Write, but do not evaluate, an expression for the average distance between the two particles on the interval $1 \leq t \leq 3$.

Part D

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$

8-1b

11. A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$



Find the average rate at which water is draining from the tank between time $t=0$ and time $t=8$ hours.

Part B

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

=258.052 or 258.053



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

=258.052 or 258.053

8-1b

12. The temperature outside a house during a 24-hour period is given by

$$F(t) = 80 - 10 \cos\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24,$$

where $F(t)$ is measured in degrees Fahrenheit and t is measured in hours.

Find the average temperature, to the nearest degree Fahrenheit, between $t = 6$ and $t = 14$.

Part B

1 point is earned for limits and $1/(14 - 6)$

1 point is earned for integrand

1 point is earned for answer

$$\text{Avg.} = \frac{1}{14-6} \int_6^{14} \left[80 - 100 \cos\left(\frac{\pi}{12}\right) \right] dt$$

$$= \frac{1}{8}(697.2957795)$$

$$= 87.162 \text{ or } 87.161$$

$$\approx 87^\circ \text{ F}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for limits and $1/(14 - 6)$

1 point is earned for integrand

1 point is earned for answer

$$\text{Avg.} = \frac{1}{14-6} \int_6^{14} \left[80 - 100 \cos\left(\frac{\pi}{12}\right) \right] dt$$

$$= \frac{1}{8}(697.2957795)$$

$$= 87.162 \text{ or } 87.161$$

$$\approx 87^\circ \text{ F}$$

8-1b

13. Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by $F(t) = 82 + 4 \sin\left(\frac{t}{2}\right)$ for $0 \leq t \leq 30$, where $F(t)$ is measured in cars per minute and t is measured in minutes.



What is the average value of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

Part C

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.999 \text{ cars/min}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.999 \text{ cars/min}$$

14. If f is continuous on the interval $[a, b]$, then there exists c such that $a < c < b$ and $\int_a^b f(x) dx =$

(A) $\frac{f(c)}{b-a}$

(B) $\frac{f(b)-f(a)}{b-a}$

(C) $f(b) - f(a)$

(D) $f'(c)(b-a)$

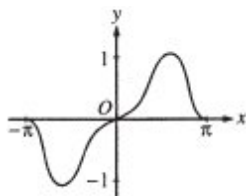
(E) $f(c)(b-a)$



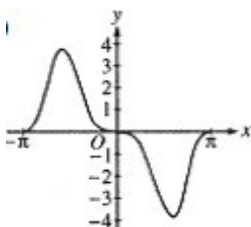
8-1b

15. The graphs of five functions are shown below. Which function has a nonzero average value over the closed interval $[-\pi, \pi]$?

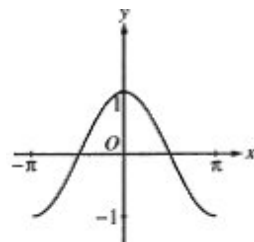
(A)



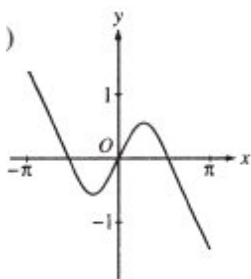
(B)



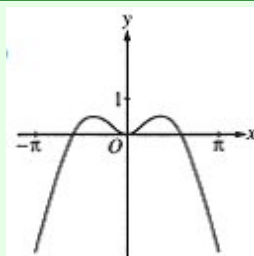
(C)



(D)



(E)



16. The average value of $1/x$ on the closed interval $[1,3]$ is

- (A) 12
(B) 23
(C) $\ln 2/2$
(D) $\ln 3/2$
(E) $\ln 3$



8-1b

17. What is the average value of y for the part of the curve $y = 3x - x^2$, which is the *first quadrant*?

(A) -6

(B) -2

(C) $\frac{3}{2}$

(D) $\frac{9}{4}$

(E) $\frac{9}{2}$



18. What is the average value of $y = x^2\sqrt{x^3 + 1}$ on the interval $[0,2]$?

(A) $\frac{26}{9}$

(B) $\frac{52}{9}$

(C) $\frac{26}{3}$

(D) $\frac{52}{3}$

(E) 24



19. The average value of $f(x) = x^2\sqrt{x^3 + 1}$ on the closed interval $[0,2]$ is

(A) $26/9$

(B) $13/3$

(C) $26/3$

(D) 13

(E) 26



20. What is the average (mean) value of $3t^3 - t^2$ over the interval $-1 \leq t \leq 2$?

(A) $11/4$

(B) $7/2$

(C) 8

(D) $33/4$

(E) 16



8-1b

21. Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f''(x) = 24x - 18$

(ii) $f'(1) = -6$

(iii) $f(2) = 0$

Find the average value of f on the interval $1 \leq x \leq 3$.

Part C

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= 5 \end{aligned}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

8-1b

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= 5 \end{aligned}$$

8-1b

22. Let f be a function such that $f'(x) = 6x + 8$.

Find the average value of $f(x)$ on the closed interval $[-1, 1]$.

Part B

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the anti differentiation and evaluation

$$\begin{aligned} & \frac{1}{1 - (-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\ & \frac{1}{2} \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_{-1}^1 \\ & = \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} + 2 \right) \right] \\ & = \boxed{-\frac{2}{3}} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the anti differentiation and evaluation

$$\begin{aligned} & \frac{1}{1 - (-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\ & \frac{1}{2} \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_{-1}^1 \\ & = \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} + 2 \right) \right] \\ & = \boxed{-\frac{2}{3}} \end{aligned}$$

8-1b

23.  What is the average value of $y = \frac{\cos x}{x^2 + x + 2}$ on the closed interval $[-1, 3]$?
- (A) -0.085
(B) 0.090
(C) 0.183 ✓
(D) 0.244
(E) 0.732
24. The average (mean) value of $\sqrt{x}$ over the interval $0 \leq x \leq 2$ is
- (A) $\frac{1}{3}\sqrt{2}$
(B) $\frac{1}{2}\sqrt{2}$
(C) $\frac{2}{3}\sqrt{2}$ ✓
(D) 1
(E) $\frac{4}{3}\sqrt{2}$
25.  The velocity, in ft/sec, of a particle moving along the x -axis is given by the function $v(t) = e^t + te^t$. What is the average velocity of the particle from time $t = 0$ to time $t = 3$?
- (A) 20.086 ft/sec ✓
(B) 26.447 ft/sec
(C) 32.809 ft/sec
(D) 40.671 ft/sec
(E) 79.342 ft/sec

8-1b

26. Let f be the function defined by $f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$

Find the average value of $f(x)$ on the closed interval $0 \leq x \leq 5$.

Part B

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5-x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + (5x - \frac{1}{2}x^2) \Big|_3^5$$

$$= (\frac{16}{3} - \frac{2}{3}) + (\frac{25}{2} - \frac{21}{2}) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

$$\text{Average value: } \frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5-x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + (5x - \frac{1}{2}x^2) \Big|_3^5$$

$$= (\frac{16}{3} - \frac{2}{3}) + (\frac{25}{2} - \frac{21}{2}) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

$$\text{Average value: } \frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$$

8-1b

27. Let f be the function defined by

$$f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$$

- (a) Is f continuous at $x = 3$? Explain why or why not.
 (b) Find the average value of $f(x)$ on the closed interval $0 \leq x \leq 5$.
 (c) Suppose the function g is defined by

$$g(x) = \begin{cases} k\sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ mx+2 & \text{for } 3 < x \leq 5, \end{cases}$$

where k and m are constants. If g is differentiable at $x = 3$, what are the values of k and m ?

Part A

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.

Part B

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

8-1b

1 point is earned for the correct antiderivative of $5 - x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value: $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5 - x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value: $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$

Part C

1 point is earned for the correct answer $2k = 3m + 2$

Since g is continuous at $x = 3$, $2k = 3m + 2$.

8-1b

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer $\frac{k}{4} = m$

Since these two limits exist and g is differentiable at $x = 3$, the two limits are equal. Thus $\frac{k}{4} = m$.

1 point is earned for the correct values for k and m

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $2k = 3m + 2$

Since g is continuous at $x = 3$, $2k = 3m + 2$.

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer $\frac{k}{4} = m$

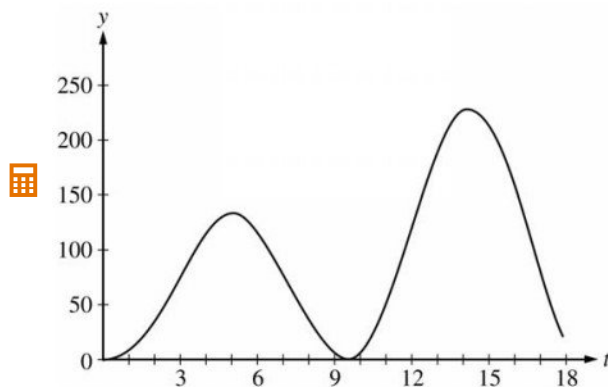
Since these two limits exist and g is differentiable at $x = 3$, the two limits are equal. Thus $\frac{k}{4} = m$.

1 point is earned for the correct values for k and m

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$

8-1b

28.



At an intersection in Thomasville, Oregon, cars turn left at the rate $L(t) = 60\sqrt{t} \sin^2\left(\frac{t}{3}\right)$ cars per hour over the time interval $0 \leq t \leq 18$ hours. The graph of $y = L(t)$ is shown above.

(a) To the nearest whole number, find the total number of cars turning left at the intersection over the time interval $0 \leq t \leq 18$ hours.

(b) Traffic engineers will consider turn restrictions when $L(t) \geq 150$ cars per hour. Find all values of t for which $L(t) \geq 150$ and compute the average value of L over this time interval. Indicate units of measure.

(c) Traffic engineers will install a signal if there is any two-hour time interval during which the product of the total number of cars turning left and the total number of oncoming cars traveling straight through the intersection is greater than 200,000. In every two-hour time interval, 500 oncoming cars travel straight through the intersection. Does this intersection require a traffic signal? Explain the reasoning that leads to your conclusion.

Part A

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\int_0^{18} L(t) dt \approx 1658 \text{ cars}$$

8-1b

Part B

1 point is earned for t -interval when $L(t) \geq 150$

1 point is earned for the average value integral

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for t -interval when $L(t) \geq 150$

1 point is earned for the average value integral

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$

Part C

1 point is earned for considering 400 cars

1 point is earned for valid interval $[h, h + 2]$

1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

8-1b

OR

1 point is earned for considers 200 cars per hour

1 point is earned for solving $L(t) \geq 200$

1 point is earned if discusses 2 hour interval

1 point is earned for the answer and explanation

For the product to exceed 200,000, the number of cars turning left in a two-hour interval must be greater than 400.

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

OR

The number of cars turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval. $L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$.

Yes, a traffic signal is required.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for considering 400 cars

1 point is earned for valid interval $[h, h + 2]$ 1 point is earned for the value of $\int_h^{h+2} L(t) dt$

1 point is earned for the answer and explanation

OR

1 point is earned for considers 200 cars per hour

1 point is earned for solving $L(t) \geq 200$

1 point is earned if discusses 2 hour interval

1 point is earned for the answer and explanation

For the product to exceed 200,000, the number of cars turning left in a two-hour interval must be greater than 400.

8-1b

$$\int_{13}^{15} L(t) dt = 431.931 > 400$$

OR

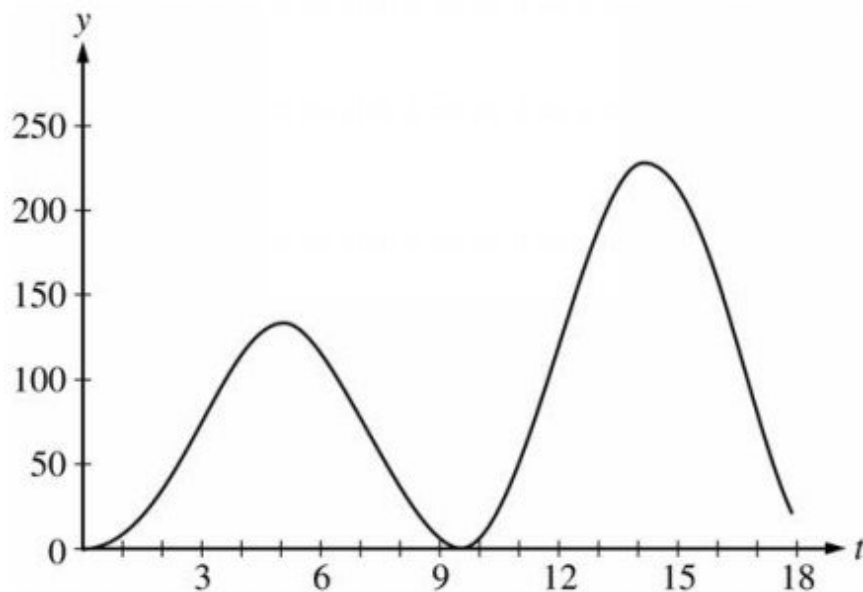
The number of cars turning left will be greater than 400 on a two-hour interval if $L(t) \geq 200$ on that interval.

$L(t) \geq 200$ on any two-hour subinterval of $[13.25304, 15.32386]$

Yes, a traffic signal is required.

8-1b

29.



At an intersection in Thomasville, Oregon, cars turn left at the rate $L(t) = 60\sqrt{t}\sin^2\left(\frac{t}{3}\right)$ cars per hour over the time interval $0 \leq t \leq 18$ hours. The graph of $y = L(t)$ is shown above.

 Traffic engineers will consider turn restrictions when $L(t) \geq 150$ cars per hour. Find all values of t for which $L(t) \geq 150$ and compute the average value of L over this time interval. Indicate units of measure.

Part B

1 point is earned for: t -interval when $L(t) \geq 150$

1 point is earned for the average value integral

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: t -interval when $L(t) \geq 150$

1 point is earned for the average value integral

8-1b

1 point is earned for the answer with units

$L(t) = 150$ when $t = 12.42831, 16.121166$

Let $R = 12.42831$ and $S = 16.121166$

$L(t) \geq 150$ for t in the interval $[R, S]$

$$\frac{1}{S-R} \int_R^S L(t) dt = 199.426 \text{ cars per hour}$$

8-1b

30.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

Based on the model B from part (c), find Bob's average velocity during the interval $0 \leq t \leq 10$.

Part D

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\
 &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\
 &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\
 &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\
 &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}
 \end{aligned}$$

8-1b

Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by $v(t) = 16 + 2 \sin(\sqrt{t+10})$ for $0 \leq t \leq 120$ minutes.

31.  The volumetric flow at a location along the river is the product of the cross-sectional area and the velocity of the water at that location. Use your approximation from part (a) to estimate the average value of the volumetric flow at Picnic Point, in cubic feet per minute, from $t = 0$ to $t = 120$ minutes.

Part B

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} \frac{115v(t)dt}{1}$$

$$=1807.169 \text{ or } 1807.170 \text{ ft}^3 / \text{min}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} \frac{115v(t)dt}{1}$$

$$=1807.169 \text{ or } 1807.170 \text{ ft}^3 / \text{min}$$

8-1b

32.  Recall that the volumetric flow is the product of the cross-sectional area and the velocity of the water at a location. To prevent flooding, water must be diverted if the average value of the volumetric flow at Picnic Point exceeds 2100 cubic feet per minute for a 20-minute period. Using your answer from part (c), find the average value of the volumetric flow during the time interval $40 \leq t \leq 60$ minutes. Does this value indicate that the water must be diverted?

Part D

1 point is earned for volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds 2100 ft³ / min



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds 2100 ft³ / min

8-1b

33. If the function f given by $f(x) = x^3$ has an average value of 9 on the closed interval $[0, k]$, then $k =$

(A) 3

(B) $3^{\frac{1}{2}}$

(C) $18^{\frac{1}{3}}$

(D) $36^{\frac{1}{4}}$

(E) $36^{\frac{1}{3}}$



8-1b

t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

34.  Use the function P defined in part (c) to find the average value, in degrees Celsius, of $P(t)$ over the time interval $0 \leq t \leq 15$ days.

Part D

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ C$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ C$$

8-1b

35. Approximate the average temperature, in degree Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^\circ C$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^\circ C$$

8-1b

36. The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.

Find the average rate of consumption of cola over the 10-year time period beginning January 1, 1983. Indicate units of measure.

Part B

1 point is earned for correctly using $[3,13]$ and dividing by $13-3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13-3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b-a} \int_a^b S(t) dt$$

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

(19.680 billion gal/yr)

$$\text{Student solves } \frac{S(13) - S(3)}{10} = S'(t) \text{ for } t_0.$$

$$S(t_0) = 19.680 \text{ billions gal/yr}$$

3/3 if correct.

Otherwise, read by standard where max 1-0-0.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for correctly using $[3,13]$ and dividing by $13-3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13-3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b-a} \int_a^b S(t) dt$$

8-1b

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

(19.680 billion gal/yr)

Student solves $\frac{S(13)-S(3)}{10} = S'(t)$ for t_0 .

$S(t_0) = 19.680$ billions gal/yr

3/3 if correct.

Otherwise, read by standard where max 1-0-0.

8-1b

37.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

The rate of water flow $R(t)$ can be approximated by $Q(t) = \frac{1}{79}(768 + 23t - t^2)$. Use $Q(t)$ to approximate the average rate of water flow during the 24-hour period. Indicate units of measure.

Part C

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\frac{1}{24} \int_0^{24} \frac{1}{79}(768 + 23t - t^2) dt$$

$$= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr}$$

1 point is earned for units

(units) Gallons in part(a) and gallons/hr in part (c), or equivalent.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value constant

8-1b

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

$$\begin{aligned} &= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt \\ &= 10.785 \text{ gal/hr, or, } 10.784 \text{ gal/hr} \end{aligned}$$

1 point is earned for units

(units) Gallons in part(a) and gallons/hr in part (c), or equivalent.

38.  If $\frac{dy}{dx} = \frac{1}{x}$, then the average rate of change of y with respect to x on the closed interval $[1,4]$ is

(A) $-\frac{1}{4}$

(B) $\frac{1}{2} \ln 2$

(C) $\frac{2}{3} \ln 2$ ✓

(D) $\frac{2}{5}$

(E) 2

8-1b

39. For $0 \leq t \leq 6$, a particle is moving along the x -axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t) = 2\sin(e^{t/4}) + 1$. The acceleration of the particle is given by $a(t) = \frac{1}{2}e^{t/4}\cos(e^{t/4})$ and $x(0) = 2$.



Find the average velocity of the particle for the time period $0 \leq t \leq 6$.

Part B

1 point is earned for integral

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$

8-1b

40. Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.



Find the time t for which the amount of grass clippings in the bin is equal to the average amount of grass clippings in the bin over the interval $0 \leq t \leq 30$.

Part C

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$

8-1b

41.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.



According to the model f , given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \leq t \leq 40$?

Part D

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

=5.916 miles per minute



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

=5.916 miles per minute

8-1b

42. A particle moves along the x -axis so that its velocity v at time t , for $0 \leq t \leq 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.



Find the average speed of the particle over the interval $0 \leq t \leq 2$.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$



0	1	2
---	---	---

The student response earns all of the following points:

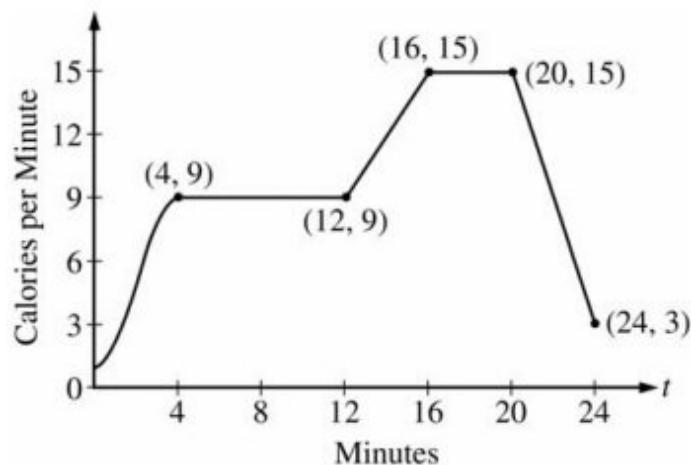
1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$

8-1b

43.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f . In the figure above, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \leq t \leq 4$ and f is piecewise linear for $4 \leq t \leq 24$.

The setting on the machine is now changed so that the person burns $f(t)+c$ calories per minute. For this setting, find c so that an average of 15 calories per minute is burned during the time interval $6 \leq t \leq 18$.

Part D

1 point is earned for the setup

1 point is earned for the value of c

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$$

This means $132+12c=15(12)$. So, $c=4$. **OR** Currently, the average is $\frac{132}{12} = 11$ calories/min. Adding c to $f(t)$ will shift the average by c . So $c=4$ to get an average of 15 calories/min.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the value of c

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$$

This means $132+12c=15(12)$. So, $c=4$.

OR Currently, the average is $\frac{132}{12} = 11$ calories/min.

8-1b

Adding c to $f(t)$ will shift the average by c . So $c=4$ to get an average of 15 calories/min.

8-1b

44.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t=0$) and 8 P.M. ($t=8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

Use a trapezoidal sum with the four subintervals given by the table to approximate the value of

$\frac{1}{8} \int_0^8 E(t) dt$. Using correct units, explain the meaning of $\frac{1}{8} \int_0^8 E(t) dt$ in terms of the number of entries.

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\frac{1}{8} \int_0^8 E(t) dt \approx$$

$$\frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right)$$

$$= 10.687 \text{ or } 10.688$$

$$\frac{1}{8} \int_0^8 E(t) dt$$

is the average number of hundreds of entries in the box between noon and 8 P.M



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

8-1b

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688 \\ \frac{1}{8} \int_0^8 E(t) dt\end{aligned}$$

is the average number of hundreds of entries in the box between noon and 8 P.M

8-1b

45.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.

Part B

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\frac{1}{8} \int_0^8 T(x) dx$:

$$A = \frac{100+93}{2} \cdot 1 + \frac{93+70}{2} \cdot 4 + \frac{70+62}{2} \cdot 1 + \frac{62+55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8} A = 75.6875^{\circ}\text{C}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

8-1b

Trapezoidal approximation for $\frac{1}{8} \int_0^8 T(x) dx$:

$$A = \frac{100+93}{2} \cdot 1 + \frac{93+70}{2} \cdot 4 + \frac{70+62}{2} \cdot 1 + \frac{62+55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8}A = 75.6875^\circ C$$

46. A particle moves along the x -axis. The velocity of the particle at time t is given by $v(t)$, and the acceleration of the particle at time t is given by $a(t)$. Which of the following gives the average velocity of the particle from time $t = 0$ to time $t = 8$?

(A) $\frac{a(8)-a(0)}{8}$

(B) $\frac{1}{8} \int_0^8 v(t) dt$



(C) $\frac{1}{8} \int_0^8 |v(t)| dt$

(D) $\frac{1}{2} \int_0^8 v(t) dt$

(E) $\frac{v(0)+v(8)}{2}$

8-1b

47.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a differentiable function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

(a) Use the data in the table to approximate $C'(3.5)$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t , $2 \leq t \leq 4$, at which $C'(t) = 2$? Justify your answer.

(c) Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\frac{1}{6} \int_0^6 C(t) dt$. Using correct units, explain the meaning of $\frac{1}{6} \int_0^6 C(t) dt$ in the context of the problem.

(d) The amount of coffee in the cup, in ounces, is modeled by $B(t) = 16 - 16e^{-0.4t}$. Using this model, find the rate at which the amount of coffee in the cup is changing when $t = 5$.

Part A

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

Part B

8-1b

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.

Part C

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

8-1b

$$\begin{aligned}\frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

$$\begin{aligned}\frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.

Part D

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$\begin{aligned}B'(t) &= -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t} \\ B'(5) &= 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}\end{aligned}$$

8-1b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$

$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$

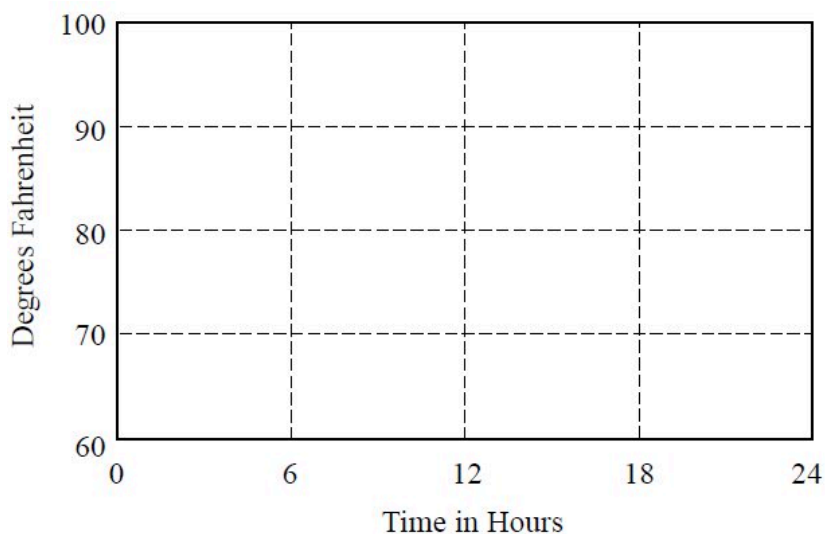
8-1b

48.  The temperature outside a house during a 24-hour period is given by

$$F(t) = 80 - 10 \cos\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24,$$

where $F(t)$ is measured in degrees Fahrenheit and t is measured in hours.

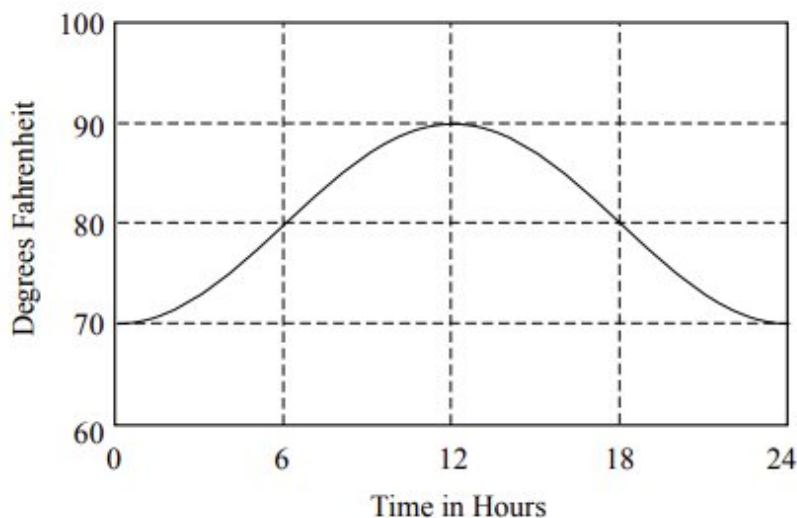
- (a) Sketch the graph of F on the grid below.



- (b) Find the average temperature, to the nearest degree Fahrenheit, between $t = 6$ and $t = 14$.
- (c) An air conditioner cooled the house whenever the outside temperature was at or above 78 degrees Fahrenheit. For what values of t was the air conditioner cooling the house?
- (d) The cost of cooling the house accumulates at the rate of \$0.05 per hour for each degree the outside temperature exceeds 78 degrees Fahrenheit. What was the total cost, to the nearest cent, to cool the house for this 24-hour period?

Part A

1 point is earned for bell-shaped graph minimum 70 at $t = 0, t = 24$ only maximum 90 at $t = 12$ only



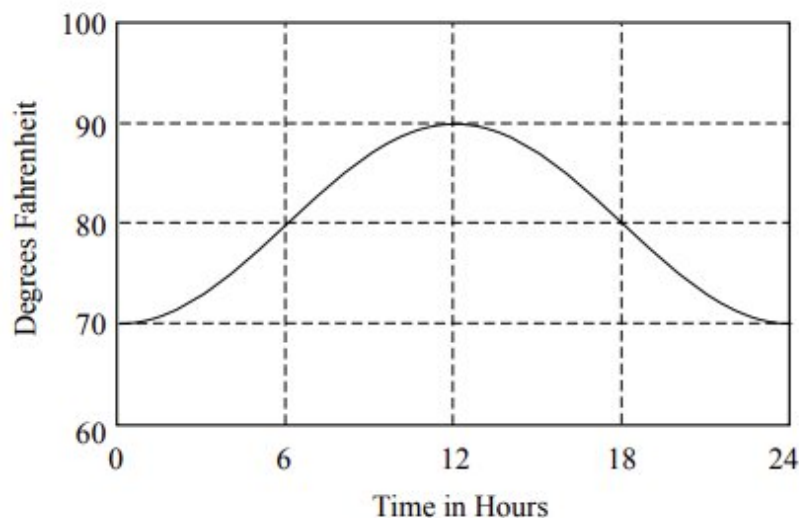
8-1b



0	1
---	---

The student response earns all of the following points:

1 point is earned for bell-shaped graph minimum 70 at $t = 0$, $t = 24$ only maximum 90 at $t = 12$ only

**Part B**

1 point is earned for limits and $1/(14 - 6)$

1 point is earned for integrand

1 point is earned for answer

$$\begin{aligned}
 \text{Avg.} &= \frac{1}{14-6} \int_6^{14} \left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] dt \\
 &= \frac{1}{8} (697.2957795) \\
 &= 87.162 \text{ or } 87.161 \\
 &= 87^\circ \text{F}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and $1/(14 - 6)$

8-1b

1 point is earned for integrand

1 point is earned for answer

$$\begin{aligned}\text{Avg.} &= \frac{1}{14-6} \int_6^{14} \left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] dt \\ &= \frac{1}{8} (697.2957795) \\ &= 87.162 \text{ or } 87.161 \\ &= 87^\circ \text{F}\end{aligned}$$

Part C

1 point is earned for inequality or equation

1 point is earned for solutions with interval

$$\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \geq 0$$

$$2 - 10 \cos \left(\frac{\pi t}{12} \right) \geq 0$$

$$\left. \begin{array}{c} 5.230 \\ \text{or} \\ 5.231 \end{array} \right\} \leq t \leq \left\{ \begin{array}{c} 18.769 \\ \text{or} \\ 18.770 \end{array} \right.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for inequality or equation

1 point is earned for solutions with interval

$$\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \geq 0$$

$$2 - 10 \cos \left(\frac{\pi t}{12} \right) \geq 0$$

$$\left. \begin{array}{c} 5.230 \\ \text{or} \\ 5.231 \end{array} \right\} \leq t \leq \left\{ \begin{array}{c} 18.769 \\ \text{or} \\ 18.770 \end{array} \right.$$

Part D

8-1b

1 point is earned for limits and 0.05

1 point is earned for integrand

1 point is earned for answer

$$\begin{array}{c}
 18.770 \\
 \text{or} \\
 18.769 \\
 C = 0.05 \int_{5.231}^{} \left(\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \right) dt \\
 \text{or} \\
 5.230 \\
 = 0.05 (101.92741) = 5.096 \approx \$5.10
 \end{array}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and 0.05

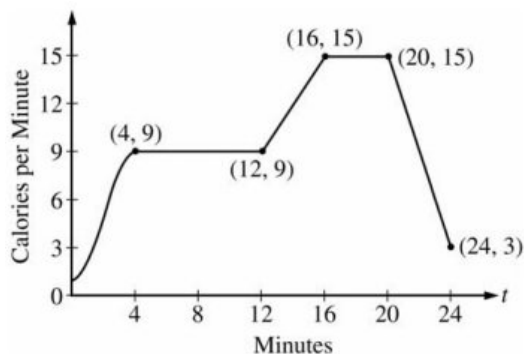
1 point is earned for integrand

1 point is earned for answer

$$\begin{array}{c}
 18.770 \\
 \text{or} \\
 18.769 \\
 C = 0.05 \int_{5.231}^{} \left(\left[80 - 10 \cos \left(\frac{\pi t}{12} \right) \right] - 78 \right) dt \\
 \text{or} \\
 5.230 \\
 = 0.05 (101.92741) = 5.096 \approx \$5.10
 \end{array}$$

8-1b

49.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f . In the figure above, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \leq t \leq 4$ and f is piecewise linear for $4 \leq t \leq 24$.

- Find $f'(22)$. Indicate units of measure.
- For the time interval $0 \leq t \leq 24$, at what time t is f increasing at its greatest rate? Show the reasoning that supports your answer.
- Find the total number of calories burned over the time interval $6 \leq t \leq 18$ minutes.
- The setting on the machine is now changed so that the person burns $f(t) + c$ calories per minute. For this setting, find c so that an average of 15 calories per minute is burned during the time interval $6 \leq t \leq 18$.

Part A

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$

Part B

1 point is earned for f' on $(0, 4)$

8-1b

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for f' on $(0, 4)$

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

8-1b

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.

Part C

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$

Part D

1 point is earned for the setup

1 point is earned for the value of c

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15.$$

This means $132 + 12c = 15(12)$. So, $c = 4$.

OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

8-1b



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for the setup

1 point is earned for the value of c

We want $\frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$.

This means $132 + 12c = 15(12)$. So, $c = 4$.

OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

8-1b

50. Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f'(x) = 24x - 18$

(ii) $f(1) = -6$

(iii) $f(2) = 0$

(a) Find each x such that the line tangent to the graph of f at $(x, f(x))$ is horizontal.

(b) Write an expression for $f(x)$.

(c) Find the average value of f on the interval $1 \leq x \leq 3$.

Part A

1 point is earned for antiderivative with or without C

1 point is earned for $f(1) = -6$ for student's $f(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for $f(1) = -6$ for student's $f(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$

8-1b

Part B

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$

$$f(2) = 0 = 32 - 36 + C$$

$$\therefore C = 4$$

$$f(x) = 4x^3 - 9x^2 + 4$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$

$$f(2) = 0 = 32 - 36 + C$$

$$\therefore C = 4$$

$$f(x) = 4x^3 - 9x^2 + 4$$

Part C

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

8-1b

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4 \, dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= \boxed{5} \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4 \, dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= \boxed{5} \end{aligned}$$

8-1b

51.  Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.
- (a) Find the average rate of change of $A(t)$ over the interval $0 \leq t \leq 30$. Indicate units of measure.
- (b) Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.
- (c) Find the time t for which the amount of grass clippings in the bin is equal to the average amount of grass clippings in the bin over the interval $0 \leq t \leq 30$.
- (d) For $t > 30$, $L(t)$, the linear approximation to A at $t = 30$, is a better model for the amount of grass clippings remaining in the bin. Use $L(t)$ to predict the time at which there will be 0.5 pound of grass clippings remaining in the bin. Show the work that leads to your answer.

Part A

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$

Part B

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $A'(15)$

8-1b

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.

Part C

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or } 12.414)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or } 12.414)$$

Part D

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

8-1b

$$A(30) = 0.782928$$

$$L(t) = 0.5 \div t = 35.054$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \div t = 35.054$$

8-1b

52. Let f be a function such that $f''(x) = 6x + 8$.

(a) Find $f(x)$ if the graph of f is tangent to the line $3x - y = 2$ at the point $(0, -2)$.

(b) Find the average value of $f(x)$ on the closed interval $[-1, 1]$.

Part A

1 point is earned for $f'(x)$ with C

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$ with C

1 point is earned for $f'(0) = 3$

1 point is earned for finding C

1 point is earned for using f' to find f with constant

1 point is earned for using $(0, -2)$ to find d

8-1b

$$f'(x) = 3x^2 + 8x + C$$

$$f'(0) = 3$$

$$C = 3$$

$$f(x) = x^3 + 4x^2 + 3x + d$$

$$d = -2$$

$$f(x) = x^3 + 4x^2 + 3x - 2$$

Part B

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the antidifferentiation and evaluation

$$\begin{aligned} & \frac{1}{1-(-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\ &= \frac{1}{2} \left[\frac{1}{4}x^4 + \frac{4}{3}x^3 + \frac{3}{2}x^2 - 2x \right]_{-1}^1 \\ &= \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} - 2 \right) \right] \\ &= \boxed{-\frac{2}{3}} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

3 points are earned for integrand, constant and limits.

2 points are **deducted** for incorrect integrand or incorrect constant

1 point is **deducted** for incorrect limits

1 point is earned for the antidifferentiation and evaluation

8-1b

$$\begin{aligned}& \frac{1}{1-(-1)} \int_{-1}^1 (x^3 + 4x^2 + 3x - 2) dx \\&= \frac{1}{2} \left[\frac{1}{4} x^4 + \frac{4}{3} x^3 + \frac{3}{2} x^2 - 2x \right]_{-1}^1 \\&= \frac{1}{2} \left[\left(\frac{1}{4} + \frac{4}{3} + \frac{3}{2} - 2 \right) - \left(\frac{1}{4} - \frac{4}{3} + \frac{3}{2} + 2 \right) \right] \\&= \boxed{-\frac{2}{3}}\end{aligned}$$

8-1b

53.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

(a) Use the data in the table to estimate the value of $v'(16)$.

(b) Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)| dt$ in the context of the problem. Approximate the value of $\int_0^{40} |v(t)| dt$ using a right Riemann sum with the four subintervals indicated in the table.

(c) Bob is riding his bicycle along the same path. For $0 \leq t \leq 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t = 5$.

(d) Based on the model B from part (c), find Bob's average velocity during the interval $0 \leq t \leq 10$.

Part A

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

8-1b

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$

Part C

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$



0	1	2
---	---	---

8-1b

The student response earns all of the following points:

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$

Part D

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min} \end{aligned}$$

8-1b

54. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15 \sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).

- How many fish enter the lake over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)? Give your answer to the nearest whole number.
- What is the average number of fish that leave the lake per hour over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)?
- At what time t , for $0 \leq t \leq 8$, is the greatest number of fish in the lake? Justify your answer.
- Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$)? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires a correct integrand and limits.

The second point requires the first point to be earned. The second point is earned with an answer of 153, 154, or any noninteger answer that is accurate to the number of places presented after the decimal point (up to the first 3). An answer between 101 and 102, inclusive, might indicate that the response uses degree mode rather than radian mode. In this case, the second point is not earned, but the remaining points for the question may be earned for consistent answers using degree mode.



0	1	2
---	---	---

8-1b

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\int_0^5 E(t) dt = 153.457690$$

To the nearest whole number, 153 fish enter the lake from midnight to 5 A.M.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a correct integrand and limits. The $\frac{1}{5}$ is considered for the second point. If this integral is computed in Part A of the question, that number can be used in place of the integral to earn the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\frac{1}{5-0} \int_0^5 L(t) dt = 6.059038$$

The average number of fish that leave the lake per hour from midnight to 5 A.M. is 6.059 fish per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned through analytical, graphical, or verbal expressions to show where $E(t) - L(t) = 0$, where $E(t) - L(t)$ changes sign, or where $E(t)$ and $L(t)$ intersect.

The second point is earned with the correct value of t .

The third point requires a global argument for the sign of $E(t) - L(t)$ rather than a local argument.

8-1b

Another approach uses the Candidates Test. Let $A(t)$ be the change in the number of fish in the lake from midnight to t hours after midnight.

$$A(t) = \int_0^t (E(s) - L(s)) \, ds$$

$$A'(t) = E(t) - L(t) = 0 \Rightarrow t = C = 6.20356$$

t	$A(t)$
0	0
C	135.01492
8	80.91908

Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ sets $E(t) - L(t) = 0$
- ☐ answer
- ☐ justification

Solution:

The rate of change in the number of fish in the lake at time t is given by $E(t) - L(t)$.

$$E(t) - L(t) = 0 \Rightarrow t = 6.20356$$

$E(t) - L(t) > 0$ for $0 \leq t < 6.20356$, and $E(t) - L(t) < 0$ for $6.20356 < t \leq 8$. Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires an attempt to use $E'(5)$ and $L'(5)$. Incorrect symbolic differentiation for both evaluated at 5 earns the point. If values are reported without symbolic work, at least one must be correct to earn the point. These values need not be reported to three places after the decimal point.

The second point requires a direct connection between $E'(5)$ and $L'(5)$. The point is not earned if reported values are incorrect. The numerical value of $E'(5) - L'(5)$ is not required, but the response must indicate that this value is negative.

8-1b



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ considers $E'(5)$ and $L'(5)$
- ☐ answer with explanation

Solution:

$$E'(5) - L'(5) = -10.7228 < 0$$

Because $E'(5) - L'(5) < 0$, the rate of change in the number of fish is decreasing at time $t = 5$.

8-1b

55. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

From 5 A.M. to 10 A.M., the rate at which vehicles arrive at a certain toll plaza is given by $A(t) = 450\sqrt{\sin(0.62t)}$, where t is the number of hours after 5 A.M. and $A(t)$ is measured in vehicles per hour. Traffic is flowing smoothly at 5 A.M. with no vehicles waiting in line.

- Write, but do not evaluate, an integral expression that gives the total number of vehicles that arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).
- Find the average value of the rate, in vehicles per hour, at which vehicles arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).
- Is the rate at which vehicles arrive at the toll plaza at 6 A.M. ($t = 1$) increasing or decreasing? Give a reason for your answer.
- A line forms whenever $A(t) \geq 400$. The number of vehicles in line at time t , for $a \leq t \leq 4$, is given by $N(t) = \int_a^t (A(x) - 400) dx$, where a is the time when a line first begins to form. To the nearest whole number, find the greatest number of vehicles in line at the toll plaza in the time interval $a \leq t \leq 4$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must be a definite integral with correct lower and upper limits to earn this point.

Because $|A(t)| = A(t)$ for $1 \leq t \leq 5$, a response of $\int_1^5 450\sqrt{\sin(0.62t)} dt$ or $\int_1^5 |A(t)| dt$ earns the point.

A response missing dt or using dx is eligible to earn the point.

A response with a copy error in the expression for $A(t)$ will earn the point only in the presence of $\int_1^5 A(t) dt$.

8-1b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

The total number of vehicles that arrive at the toll plaza from 6 a.m. to 10 a.m. is given by $\int_1^5 A(t) dt$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of the average value formula, indicating that $a = 1$ and $b = 5$, can be presented in single or multiple steps to earn the first point. For example, the following response earns both points: $\int_1^5 A(t) dt = 1502.147865$, so the average value is 375.536966 .

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$\int_1^5 A(t) dt = 1502.147865 = 375.536966$. so the average value is 375.536966 .

The answer must be correct to three decimal places. For example, $\frac{1}{5-1} \int_1^5 A(t) dt = 375.536966 \approx 376$ earns only the first point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode,

$\frac{1}{4} \int_1^5 A(t) dt = 79.416068$.

Special case: $\frac{1}{5} \int_1^5 A(t) dt = 300.429573$ earns 1 out of 2 points.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Uses average value formula: $\frac{1}{b-a} \int_a^b A(t) dt$

☐ Answer

8-1b

Solution:

$$\text{Average} = \frac{1}{5-1} \int_1^5 A(t) dt = 375.536966$$

The average rate at which vehicles arrive at the toll plaza from 6 a.m. to 10 a.m. is 375.537 (or 375.536) vehicles per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $A'(1)$. The second line of the model solution earns both points.

An incorrect value assigned to $A'(1)$ earns the first point (but will not earn the second point).

Without a reference to $t = 1$, the first point is earned by any of the following:

- 148.947 accurate to the number of decimals presented, with zero up to three decimal places (i.e., 149, 148, 148.9, 148.95, or 148.94)
- $A'(t) = 148.947$ by itself

To be eligible for the second point, the first point must be earned.

To earn the second point, there must be a reference to $t = 1$.

Degree mode: $A'(1) = 23.404311$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $A'(1)$
- ☐ Answer with reason

Solution:

$$A'(1) = 148.947272$$

Because $A'(1) > 0$, the rate at which the vehicles arrive at the toll plaza is increasing.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-1b

It is not necessary to indicate that $A(t) = 400$ to earn the first point, although this statement alone would earn the first point.

A response of “ $A(t) \geq 400$ when $1.469372 \leq t \leq 3.597713$ ” will earn the first 2 points. A response of “ $A(t) \geq 400$ ” along with the presence of exactly one of the two numbers above will earn the first point, but not the second. A response of “ $A(t) \geq 400$ ” by itself will not earn either of the first 2 points.

To earn the second point the values for a and b must be accurate to the number of decimals presented, with at least one and up to three decimal places. These may appear only in a candidates table, as limits of integration, or on a number line.

A response with incorrect notation involving t or x is eligible to earn all 4 points.

A response that does not earn the first point is still eligible for the remaining 3 points.

To earn the third point, a response must present the greatest number of vehicles. This point is earned for answers of either 71 or 71.254*** only.

A correct justification earns the fourth point, even if the third point is not earned because of a decimal presentation error.

When using a Candidates Test, the response must include the values for $N(a)$, $N(b)$, and $N(4)$ to earn the fourth point. These values must be correct to the number of decimals presented, with up to three decimal places. (Correctly rounded integer values are acceptable.)

Alternate solution for the third and fourth points:

For $a \leq t \leq b$, $A(t) \geq 400$. For $b \leq t \leq 4$, $A(t) \leq 400$.

Thus, $N(t) = \int_a^t (A(x) - 400)dx$ is greatest at $t = b$.

$N(b) = 71.254129$, and the greatest number of vehicles in line is 71.

Degree mode: The response is only eligible to earn the first point because in degree mode $A(t) < 400$.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Considers $N'(t) = 0$
- ☐ $t = a$ and $t = b$
- ☐ Answer
- ☐ Justification

8-1b**Solution:**

$$N'(t) = A(t) - 400 = 0$$

$$\Rightarrow A(t) = 400 \Rightarrow t = 1.469372, t = 3.597713$$

$$a = 1.469372$$

$$b = 3.597713$$

t	$N(t) = \int_a^t (A(x) - 400) dx$
a	0
b	71.254129
4	62.338346

The greatest number of vehicles in line is 71.

8-1b

56. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	2	4	6	8	10	12
$R(t)$ (vehicles per hour)	2935	3653	3442	3010	3604	1986	2201

On a certain weekday, the rate at which vehicles cross a bridge is modeled by the differentiable function R for $0 \leq t \leq 12$, where $R(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). Values of $R(t)$ for selected values of t are given in the table above.

Use the data in the table to approximate $R'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $R'(5)$ in the context of the problem.

Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\int_0^{12} R(t) \, dt$. Indicate units of measure.

On a certain weekend day, the rate at which vehicles cross the bridge is modeled by the function H defined by $H(t) = -t^3 - 3t^2 + 288t + 1300$ for $0 \leq t \leq 17$, where $H(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). According to this model, what is the average number of vehicles crossing the bridge per hour on the weekend day for $0 \leq t \leq 12$?

For $12 < t < 17$, $L(t)$, the local linear approximation to the function H given in part (c) at $t = 12$, is a better model for the rate at which vehicles cross the bridge on the weekend day. Use $L(t)$ to find the time t , for $12 < t < 17$, at which the rate of vehicles crossing the bridge is 2000 vehicles per hour. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.

8-1b

The second point requires a time reference, a rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ approximation using values from table
- ☐ interpretation with units

Solution:

$$R'(5) \approx \frac{R(6) - R(4)}{6 - 4} = \frac{3010 - 3442}{2} = -216$$

At time $t = 5$ hours (12 p.m.), the rate at which vehicles cross the bridge is decreasing at a rate of approximately 216 vehicles per hour per hour.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

- The first point does not require a simplified answer. Substitution of function values is required. A single substitution error for one and only one of $R(2)$, $R(6)$, or $R(10)$ will still earn this point.

The second point requires earning the first point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ midpoint sum setup
- ☐ approximation using values from table with units

Solution:

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx 4(R(2) + R(6) + R(10)) \\ &= 4(3653 + 3010 + 1986) \\ &= 34,596 \text{ vehicles} \end{aligned}$$

8-1b

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct integrand and limits of integration. The constant $\frac{1}{12}$ is not needed to earn this point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\frac{1}{12-0} \int_0^{12} H(t) dt = 2452$$

On the weekend day for $0 \leq t \leq 12$, the average number of vehicles crossing the bridge is 2452 vehicles per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ slope
- ☐ $L(t) = 2000$
- ☐ answer

Solution:

$$L(t) = H(12) + H'(12)(t - 12)$$

$$H(12) = 2596$$

$$H'(12) = -216$$

8-1b

$$L(t) = 2000 \Rightarrow t = 14.759259$$

On the weekend day, the rate at which vehicles cross the bridge is 2000 vehicles per hour at time $t = 14.759$ hours.

8-1b

57. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

r (centimeters)	0	1	2	2.5	4
$f(r)$ (milligrams per square centimeter)	1	2	6	10	18

The density of a bacteria population in a circular petri dish at a distance r centimeters from the center of the dish is given by an increasing, differentiable function f , where $f(r)$ is measured in milligrams per square centimeter. Values of $f(r)$ for selected values of r are given in the table above.

(a) Use the data in the table to estimate $f'(2.25)$. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) The total mass, in milligrams, of bacteria in the petri dish is given by the integral expression

$2\pi \int_0^4 r f(r) \, dr$. Approximate the value of $2\pi \int_0^4 r f(r) \, dr$ using a right Riemann sum with the four subintervals indicated by the data in the table.

(c) Is the approximation found in part (b) an overestimate or underestimate of the total mass of bacteria in the petri dish? Explain your reasoning.

(d) The density of bacteria in the petri dish, for $1 \leq r \leq 4$, is modeled by the function g defined by $g(r) = 2 - 16(\cos(1.57\sqrt{r}))^3$. For what value of k , $1 < k < 4$, is $g(k)$ equal to the average value of $g(r)$ on the interval $1 \leq r \leq 4$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-1b

To earn the first point the response must provide both a difference and a quotient and must explicitly use values of f from the table.

Simplification of the numerical value is not required to earn the first point. If the numerical value is simplified, it must be correct.

The interpretation requires all of the following: distance $r = 2.25$, density of bacteria (population) is increasing or changing, at a rate of 8, and units of milligrams per square centimeter per centimeter.

The second point (interpretation) cannot be earned without a nonzero presented value for $f'(2.25)$.

To earn the second point the interpretation must be consistent with the presented nonzero value for $f'(2.25)$. In particular, if the presented value for $f'(2.25)$ is negative, the interpretation must include “decreasing at a rate of $|f'(2.25)|$ ” or “changing at a rate of $f'(2.25)$.” The second point cannot be earned for an incorrect statement such as “the bacteria density is decreasing at a rate of $-8\dots$ ” even for a presented $f'(2.25) = -8$.

The units ($\text{mg}/\text{cm}^2/\text{cm}$) may be attached to the estimate of $f'(2.25)$ and, if so, do not need to be repeated in the interpretation.

If units attached to the estimate do not agree with units in the interpretation, read the units in the interpretation.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Estimate
- ☐ Interpretation with units

Solution:

$$f'(2.25) \approx \frac{f(2.5) - f(2)}{2.5 - 2} = \frac{10 - 6}{0.5} = 8$$

At a distance of $r = 2.25$ centimeters from the center of the petri dish, the density of the bacteria population is increasing at a rate of 8 milligrams per square centimeter per centimeter.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The presence or absence of 2π has no bearing on earning the first point.

The first point is earned for a sum of four products with at most one error in any single value among the four products. Multiplication by 1 in any term does not need to be shown, but all other products must be explicitly shown.

8-1b

A response of $1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)$ earns the first point but not the second.

A response with any error in the Riemann sum is not eligible for the second point.

A response that provides a completely correct left Riemann sum for $2\pi \int_0^4 r f(r) dr$ and approximation (91π) earns one of the two points. A response that has any error in a left Riemann sum or evaluation for $2\pi \int_0^4 r f(r) dr$ earns no points.

A response that provides a completely correct right Riemann sum for $2\pi \int_0^4 f(r) dr$ and approximation (80π) earns one of the two points. A response that has any error in a right Riemann sum or evaluation for $2\pi \int_0^4 f(r) dr$ earns no points.

Simplification of the numerical value is not required to earn the second point. If a numerical value is given, it must be correct to three decimal places.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation

Solution:

$$\begin{aligned}
 2\pi \int_0^4 r f(r) dr &\approx 2\pi(1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) \\
 &\quad + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)) \\
 &= 2\pi(1 \cdot 2 \cdot 1 + 2 \cdot 6 \cdot 1 + 2.5 \cdot 10 \cdot 0.5 + 4 \cdot 18 \cdot 1.5) \\
 &= 269\pi = 845.088
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point a response must explain that $r f(r)$ is increasing and, therefore, the right Riemann sum is an overestimate. The second point can be earned without having earned the first point.

A response that attempts to explain based on a left Riemann sum for $2\pi \int_0^4 r f(r) dr$ from part (b) earns no points.

8-1b

A response that attempts to explain based on a right Riemann sum for $2\pi \int_0^4 f(r) \, dr$ from part (b) earns no points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Product rule expression for $\frac{d}{dr}(rf(r))$
- ☐ Answer with explanation

Solution:

$$\frac{d}{dr}(rf(r)) = f(r) + rf'(r)$$

Because f is nonnegative and increasing, $\frac{d}{dr}(rf(r)) > 0$ on the interval $0 \leq r \leq 4$. Thus, the integrand $rf(r)$ is strictly increasing.

Therefore, the right Riemann sum approximation of $2\pi \int_0^4 rf(r) \, dr$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral, with or without $\frac{1}{4-1}$ or $\frac{1}{3}$.

A response that presents a definite integral with incorrect limits but a correct integrand earns the first point.

Presentation of the numerical value 9.875795 is not required to earn the second point. This point can be earned by the average value setup: $\frac{1}{3} \int_1^4 g(r) \, dr$.

Once earned for the average value setup, the second point cannot be lost. Subsequent errors will result in not earning the third point.

The third point is earned only for the value $k = 2.497$.

The third point cannot be earned without the second.

Special case: A response that does not provide the average value setup but presents an average value of -13.955 is using degree mode on their calculator. This response would not earn the second point but could earn the third point for an answer of $k = 2.5$ (or 2.499).

8-1b



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Definite integral
- ☐ Average value
- ☐ Answer

Solution:

$$\text{Average value} = g_{\text{avg}} = \frac{1}{4-1} \int_1^4 g(r) \, dr$$

$$\frac{1}{4-1} \int_1^4 g(r) \, dr = 9.875795$$

$$g(k) = g_{\text{avg}} \Rightarrow k = 2.497$$

8-1b

58. Two particles move along the x -axis. For $0 \leq t \leq 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.

- (a) For $0 \leq t \leq 6$, find all times t during which particle R is moving to the right.
- (b) For $0 \leq t \leq 6$, find all times t during which the two particles travel in opposite directions.
- (c) Find the acceleration of particle P at time $t = 3$. Is particle P speeding up, slowing down, or doing neither at time $t = 3$? Explain your reasoning.
- (d) Write, but do not evaluate, an expression for the average distance between the two particles on the interval $1 \leq t \leq 3$.

Part A

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.



0	1	2
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The student response earns all of the following points:

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.

8-1b

Part B

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(t) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(t) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.

Part C

1 point is earned for $p''(3)$

1 point is earned for answer with reason

8-1b

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $p''(3)$

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t = 3$.

Part D

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

8-1b

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$

8-1b

59.  A particle moves along the x -axis so that its velocity v at time t , for $0 \leq t \leq 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

- (a) Find the acceleration of the particle at time $t = 4$.
- (b) Find all times t in the open interval $0 < t < 5$ at which the particle changes direction. During which time intervals, for $0 \leq t \leq 5$, does the particle travel to the left?
- (c) Find the position of the particle at time $t = 2$.
- (d) Find the average speed of the particle over the interval $0 \leq t \leq 2$.

Part A

1 point is earned for the answer

$$a(4) = v'(4) = \frac{5}{7}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$a(4) = v'(4) = \frac{5}{7}$$

Part B

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$$v(t) = 0$$

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t - 2)(t - 1) = 0$$

$$t = 1, 2$$

$$v(t) > 0 \text{ for } 0 < t < 1$$

$$v(t) < 0 \text{ for } 1 < t < 2$$

8-1b

$$v(t) > 0 \text{ for } 2 < t < 5$$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$$v(t) = 0$$

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t - 2)(t - 1) = 0$$

$$t = 1, 2$$

$$v(t) > 0 \text{ for } 0 < t < 1$$

$$v(t) < 0 \text{ for } 1 < t < 2$$

$$v(t) > 0 \text{ for } 2 < t < 5$$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

Part C

1 point is earned for $\int_0^2 \ln(u^2 - 3u + 3) du$

1 point is earned if handles initial condition

1 point is earned for answer

8-1b

$$s(t) = s(0) + \int_0^t \ln(u^2 - 3u + 3) du$$

$$s(2) = 8 + \int_0^2 \ln(u^2 - 3u + 3) du$$

$$= 8.368 \text{ or } 8.369$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln(u^2 - 3u + 3) du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t) = s(0) + \int_0^t \ln(u^2 - 3u + 3) du$$

$$s(2) = 8 + \int_0^2 \ln(u^2 - 3u + 3) du$$

$$= 8.368 \text{ or } 8.369$$

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$



0	1	2
---	---	---

The student response earns all of the following points:

8-1b

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$

8-1b

60.  For $0 \leq t \leq 6$, a particle is moving along the x -axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t) = 2\sin(e^{t/4}) + 1$. The acceleration of the particle is given by $a(t) = 1/2e^{t/4}\cos(e^{t/4})$ and $x(0) = 2$.

- (a) Is the speed of the particle increasing or decreasing at time $t = 5.5$? Give a reason for your answer.
 (b) Find the average velocity of the particle for the time period $0 \leq t \leq 6$.
 (c) Find the total distance traveled by the particle from time $t = 0$ to $t = 6$.
 (d) For $0 \leq t \leq 6$, the particle changes direction exactly once. Find the position of the particle at that time.

Part A

2 points are earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.

Part B

1 point is earned for integral

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

8-1b

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)| dt = 12.573$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)| dt = 12.573$$

Part D

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-1b

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

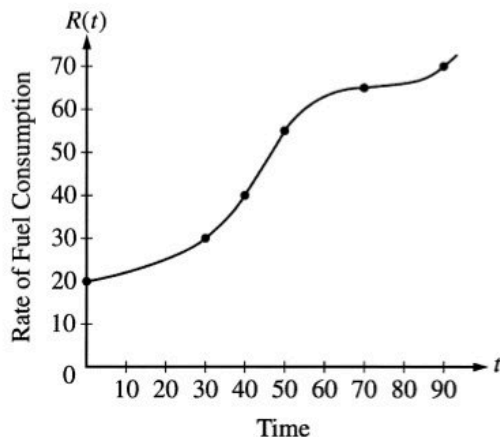
$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$

8-1b

61.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

(a) Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

(b) The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

(c) Approximate the value of $\int_0^{90} R(t) dt$ using a left Riemann sum with the five subintervals indicated by the data in the table. Is this numerical approximation less than the value of $\int_0^{90} R(t) dt$? Explain your reasoning.

(d) For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane.

Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units



0	1	2
---	---	---

The student response earns all of the following points:

8-1b

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

Part C

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct value of left Riemann sum

8-1b

$$\int_0^{90} R(t)dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.

Part D

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

8-1b

<-1> if no reference to time b

1 point is earned for the correct units in both answers

8-1b

62.  A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

- (a) Is r continuous at $t = 5$? Show the work that leads to your answer.
 (b) Find the average rate at which water is draining from the tank between time $t = 0$ and time $t = 8$ hours.
 (c) Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.
 (d) Write, but do not solve, an equation involving an integral to find the time A when the amount of water in the tank is 9000 liters.

Part A

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.

Part B

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

8-1b

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$

Part C

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

8-1b

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².

Part D

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$

8-1b

63.

 Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by

$$v(t) = 16 + 2 \sin(\sqrt{t+10}) \text{ for } 0 \leq t \leq 120 \text{ minutes.}$$

- (a) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate the area of the cross section of the river at Picnic Point, in square feet. Show the computations that lead to your answer.
- (b) The volumetric flow at a location along the river is the product of the cross-sectional area and the velocity of the water at that location. Use your approximation from part (a) to estimate the average value of the volumetric flow at Picnic Point, in cubic feet per minute, from $t = 0$ to $t = 120$ minutes.
- (c) The scientist proposes the function f , given by $f(x) = 8\sin(\pi x/24)$, as a model for the depth of the water, in feet, at Picnic Point x feet from the river's edge. Find the area of the cross section of the river at Picnic Point based on this model.
- (d) Recall that the volumetric flow is the product of the cross-sectional area and the velocity of the water at a location. To prevent flooding, water must be diverted if the average value of the volumetric flow at Picnic Point exceeds 2100 cubic feet per minute for a 20-minute period. Using your answer from part (c), find the average value of the volumetric flow during the time interval $40 \leq t \leq 60$ minutes. Does this value indicate that the water must be diverted?

Part A

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$= 115 \text{ ft}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for trapezoidal approximation

$$\frac{(0+7)}{2} \cdot 8 + \frac{(7+8)}{2} \cdot 6 + \frac{(8+2)}{2} \cdot 8 + \frac{(2+0)}{2} \cdot 2$$

$$= 115 \text{ ft}^2$$

8-1b

Part B

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$

Part C

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

8-1b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

Part D

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3/\text{min}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

8-1b

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3 / \text{min}$.

8-1b

64.



t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

(a) Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate $\int_0^{40} v(t) dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of $\int_0^{40} v(t) dt$ in terms of the plane's flight.

(b) Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

(c) The function f , defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3 \sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \leq t \leq 40$. According to this model, what is the acceleration of the plane at $t = 23$? Indicate units of measure.

(d) According to the model f , given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \leq t \leq 40$?

Part A

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is

8-1b

$$10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

Part B

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .

Part C

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$

Part D

8-1b

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute

8-1b

65.  Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by

$$F(t) = 82 + 4 \sin\left(\frac{t}{2}\right) \text{ for } 0 \leq t \leq 30,$$

where $F(t)$ is measured in cars per minute and t is measured in minutes.

(a) To the nearest whole number, how many cars pass through the intersection over the 30-minute period?

(b) Is the traffic flow increasing or decreasing at $t = 7$? Give a reason for your answer.

(c) What is the average value of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

(d) What is the average rate of change of the traffic flow over the time interval $10 \leq t \leq 15$? Indicate units of measure.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t) dt = 2474 \text{ cars}$$

Part B

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

8-1b

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

$$F'(7) = -1.872 \text{ or } -1.873$$

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

Part C

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5} \int_{10}^{15} F(t) dt = 81.899 \text{ cars/min}$$

8-1b

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

Part D

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \text{ cars/min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min² in (d)

8-1b

66.

t (hours)	$R(t)$ (gallons per hour)
0	9.6
3	10.4
6	10.8
9	11.2
12	11.4
15	11.3
18	10.7
21	10.2
24	9.6

The rate at which water flows out of a pipe, in gallons per hour, is given by a differentiable function R of time t . The table above shows the rate as measured every 3 hours for a 24-hour period.

(a) Use a midpoint Riemann sum with 4 subdivisions of equal length to approximate $\int_0^{24} R(t)dt$. Using correct units, explain the meaning of your answer in terms of water flow.

(b) Is there some time t , $0 < t < 24$, such that $R'(t) = 0$? Justify your answer.

(c) The rate of water flow $R(t)$ can be approximated by $Q(t) = \frac{1}{79}(768 + 23t - t^2)$. Use $Q(t)$ to approximate the average rate of water flow during the 24-hour time period. Indicate units of measure.

Part A

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

1 point is earned for the answer

1 point is earned for the explanation

$$\begin{aligned}
 \int_0^{24} R(t)dt &\approx 6[R(3) + R(9) + R(15) + R(21)] \\
 &= 6[10.4 + 11.2 + 11.3 + 10.2] \\
 &= 258.6 \text{ gallons}
 \end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $R(3) + R(9) + R(15) + R(21)$

8-1b

1 point is earned for the answer

1 point is earned for the explanation

$$\begin{aligned}\int_0^{24} R(t) dt &\approx 6 [R(3) + R(9) + R(15) + R(21)] \\ &= 6 [10.4 + 11.2 + 11.3 + 10.2] \\ &= 258.6 \text{ gallons}\end{aligned}$$

This is an approximation to the total flow in gallons of water from the pipe in the 24-hour period.

Part B

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for MVT or equivalent

Yes;

Since $R(0) = R(24) = 9.6$, the Mean Value Theorem guarantees that there is a t , $0 < t < 24$, such that $R'(t) = 0$.

Part C

1 point is earned for the limits and average value constant

1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

8-1b

$\approx$ average value of $Q(t)$

$$= \frac{1}{24} \int_0^{24} \frac{1}{79} (768 + 23t - t^2) dt$$

$$= 10.785 \text{ gal/hr or } 10.784 \text{ gal/hr}$$

1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.



0	1	2	3	4
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The student response earns all of the following points:

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1 point is earned for: $Q(t)$ as integrand

1 point is earned for the answer

Average rate of flow

$\approx$ average value of $Q(t)$

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1 point is earned for units

(units) Gallons in part (a) and gallons/hr in part (c), or equivalent.

8-1b

67.



t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

- (a) Use data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.
- (b) Approximate the average temperature, in degrees Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.
- (c) A student proposes the function P , given by $P(t) = 20 + 10te^{(-t/3)}$, as a model for the temperature of the water in the pond at time t , where t is measured in days and $P(t)$ is measured in degrees Celsius. Find $P'(12)$. Using appropriate units, explain the meaning of your answer in terms of water temperature.
- (d) Use the function P defined in part (c) to find the average value, in degrees Celsius, of $P(t)$ over the time interval $0 \leq t \leq 15$ days.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

8-1b

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$

Part C

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^{\circ}\text{C/day} \end{aligned}$$

8-1b

This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C}/\text{day}$ when $t = 12$ days.



0	1	2
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The student response earns all of the following points:

1 point is earned for $P'(12)$ (with or without units)

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This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C}/\text{day}$ when $t = 12$ days.

Part D

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^{\circ}\text{C}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

8-1b

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^{\circ}\text{C}$$